Inference-time proofreader guidance

Can the released proofreader improve contact maps by filtering proposed contacts, using future evidence, searching several prefixes, repairing errors, or combining independent rollout pools?

Completed validation supports larger ordinary pools and a small benefit from weighting votes by predicted rollout quality. Forced long guided rollouts expose a substantial overconfidence failure. The original 15-policy pilot is still running; final policy selection and its new reserved-test comparison remain pending.

Frozen models and experimental references

Generator: exp277 step 266344. Proofreader: exp356 step 18000, with a frozen Qwen3 causal backbone and a four-layer, 512-wide bidirectional reader containing 22.18 million trainable parameters. It estimates contact correctness and rollout recall. This experiment changes inference, not model weights.

Reader training used 50,000 experimental PDB chain structures. Reference contacts inherit exp222 native-only pyconfind labels: 3 Å side-chain threshold, minimum contact degree 0.001, sequence separation at least six. Complete contact sets and the frozen contiguous-sequence preparation rules determine recall.

Pilot, validation and reserved comparison

The pilot fixes 32 validation proteins from 32 related groups, spanning 33–937 residues. Representatives and length ranks were chosen without model-quality selection. Four rollouts for each of 15 policies give 1,920 guided trajectories.

The larger validation panel excludes every pilot group: 1,375 structures in 255 groups, with 1,371 contributing long-range metrics. H100 and GB200 ordinary pool experiments use this same biological cohort and separate runtime samples. The reserved cohort contains 1,473 structures in 295 separate related groups. Its new comparison is restricted to one frozen pair after validation selection.

What top ten percent means

At each boundary, draw K complete candidate contacts from the grammar-constrained generator. Score each hypothetical appended prefix with the reader. Uniformly sample among the highest ceil(0.1 K) proposal slots: two of 16, or seven of 64. Repeated proposals retain their sampled multiplicity. This does not enumerate the entire residue-pair universe or guarantee 90% contact accuracy.

Soft guidance instead weights proposals by q to a fixed power. The generator's probability already enters through proposal sampling. Committed pairs cannot repeat; valid pairs have sequence separation at least six. The generator samples EOS once per boundary, without intervention by the proofreader.

Other search policies

Lookahead scores a candidate after two separate futures of eight or 32 contacts, then commits only that candidate. Total-score selection lets a proposal change the reader's judgments of all earlier contacts. Eight-particle search compares prefix F1 estimates with a Jaccard diversity penalty after blocks of eight.

Repair identifies the lowest-confidence contact in the first half of a rollout, rewinds to it, and regenerates the suffix. Ordinary and uniform-proposal controls isolate guidance effects. All 15 original families must finish before selection; at most three promising new configurations may advance to larger validation.

Metrics and compute accounting

Primary quality is all-contact R-precision, with long-range R-precision requiring separation at least 24. R is the native contact count and is used only for evaluation. Deployable budgets select L, L/2 or L/5 contacts from the ranked map. Full-universe AP includes unproposed pairs at score zero; missing prediction slots count as misses. No native label sets a sampling or ranking decision.

Paired 95% intervals resample related groups and retain equal structure weights. GPU-stage timings include discarded proposals, lookahead and final scoring. They exclude staging, loading, CPU aggregation, queueing and preempted work. H100 and GB200 times remain separate. These are measured implementation costs, not hardware-independent lower bounds on guidance cost or billed GPU-hours.

Ordinary sampling: a strong practical baseline

On all 1,375 larger-validation structures, increasing ordinary frequency-ranked pools from eight to 128 improves R-precision from 0.50137 to 0.54769. The paired gain is 0.0463 (95% group interval 0.0403–0.0526); long-range R-precision rises from 0.47089 to 0.52153. Candidate coverage reaches 95.23%.

Mean warm generation cost rises from 2.431 to 6.599 H100 seconds per map. GB200 replicates the quality result: 0.50086 to 0.54779, at 1.196 and 3.054 seconds. The runtime differs as well as hardware, so this is not a controlled measurement of GPU speed alone. Single-batch ordinary pools are the preferred practical cost baseline; staged nested pools remain a separate experiment.

Contact confidence helps small pools

On the full H100 validation panel, sum-q ranking improves eight-rollout R-precision from 0.50137 to 0.51737: paired gain 0.01601 (95% interval 0.01239–0.02033). Long-range gain is 0.01710 (0.01240–0.02223), at 1.061 times the measured warm cost.

At 128 rollouts, sum q changes R-precision by −0.00008 (−0.00386–0.00424). It replaces 8.70% of top-R slots, but correct contacts gained and lost nearly cancel; exact swap counts reproduce every protein's metric change. Mean confidence alone performs poorly because it discards the vote-count signal.

Rollout-quality weighting gives a smaller validation gain

At 128 rollouts, each trajectory can weight its eligible contact votes by predicted F1, computed from its predicted precision and recall without labels. On H100, this gives R-precision 0.55195 versus 0.54769 for frequency: paired gain 0.00426 (95% interval 0.00173–0.00763). Long-range gain is 0.00412 (0.00129–0.00781). Mean warm time is 8.921 versus 6.599 seconds.

GB200 replicates the gain on the same proteins: +0.00459 (0.00207–0.00790) overall and +0.00453 (0.00164–0.00813) long-range, at 4.622 versus 3.054 seconds. The rule was added after the ordinary-pool pilot. These unadjusted validation intervals do not replace the pending reserved-test confirmation.

Larger pools saturate the top-R map

On the complete 32-protein H100 pilot, frequency R-precision is 0.60365, 0.60276 and 0.60299 at 128, 512 and 2,048 rollouts. The 2,048-minus-128 difference is −0.00066 (95% interval −0.00489–0.00352), at 15.35 times the warm cost. Coverage reaches 99.21%, and AP improves by 0.01334.

The GB200 frequency and scored-pool checks agree on the plateau. With F1-weighted votes, 2,048 versus 128 changes R-precision by +0.00002 (−0.00463–0.00500), at 20.83 times the cost. AP improves by 0.00897 (0.00480–0.01397). More candidates improve recovery below the top-R cutoff; these experiments do not establish a better top-R map.

Forced length: better individual precision, weaker maps

An adaptive control disables EOS for ordinary and top-10%-of-16 generation, evaluating exactly L and 2L contacts. All 256 continuations and their 512 nested prefix maps are complete on the pilot. The same continuation supplies both budgets; they are not independent observations.

At L, filtering raises mean individual precision from 0.43345 to 0.49286, but four-rollout map R-precision is 0.57084 versus 0.57152 for control, at 3.25 times the cost. At 2L it falls from 0.56116 to 0.53276: difference −0.02839 (95% interval −0.04961 to −0.00827), at 4.17 times the cost. Guided 2L precision is 0.28286 while the reader predicts 0.52488.

Later contacts can corrupt earlier judgments

In guided fixed-length continuations, the new second L contacts have actual precision 0.07286 while predicted correctness averages 0.50726. Control values are 0.12232 and 0.24450. Both sets use all 32 proteins and four trajectories.

The identical first L contacts are also rescored too confidently after continuing: predicted correctness rises from 0.49471 to 0.54250 while actual precision remains 0.49286. Brier score worsens by 0.02068 (95% interval 0.01511–0.02654) and AUROC falls by 0.04665 (−0.06241 to −0.03346). Later evidence is unreliable under this forced guided distribution. Distribution shift and score exploitation are consistent with these results, but a unique causal mechanism is not identified.

Bounded refinement of rollout weights

A small adaptive follow-up fixed F1 exponents 1, 2, 4 and 8 before computing their outcomes on the complete pilot. It reuses identical 128-rollout pools and reader outputs without new GPU inference. Original frequency and power-1 results reproduce exactly for every input.

Every paired all-contact and long-range interval versus exponent 1 includes zero. Power 2 changes all-contact R-precision by +0.00024 (−0.00204–0.00259) on H100 and +0.00137 (−0.00181–0.00584) on GB200. This pilot does not establish a stronger weighting rule. It does not select a final policy before the original guided pilot completes.

Numerical validation and recoverability

All frozen backbone tensors match the generator. H100 BF16 incremental scoring failed equivalence checks, so guidance uses verified FP32 caches and final maps use released BF16 full-forward scoring. GB200 full-forward replay passes with PyTorch 2.10.0+cu128; cached guidance fails the original tolerance and is excluded. No tolerance was relaxed. Eleven CPU tests cover numerical and analysis behavior.

Owned jobs use the existing CoreWeave batch queues. Source bundles, models, plans and seeds are pinned; complete trajectories commit durably to S3. Controller outages and preemptions required recovery. An early repeated-staging incident is disclosed in the execution appendix. Later regional mirrors enforce durable transfer reservations. No shared cluster configuration was changed.

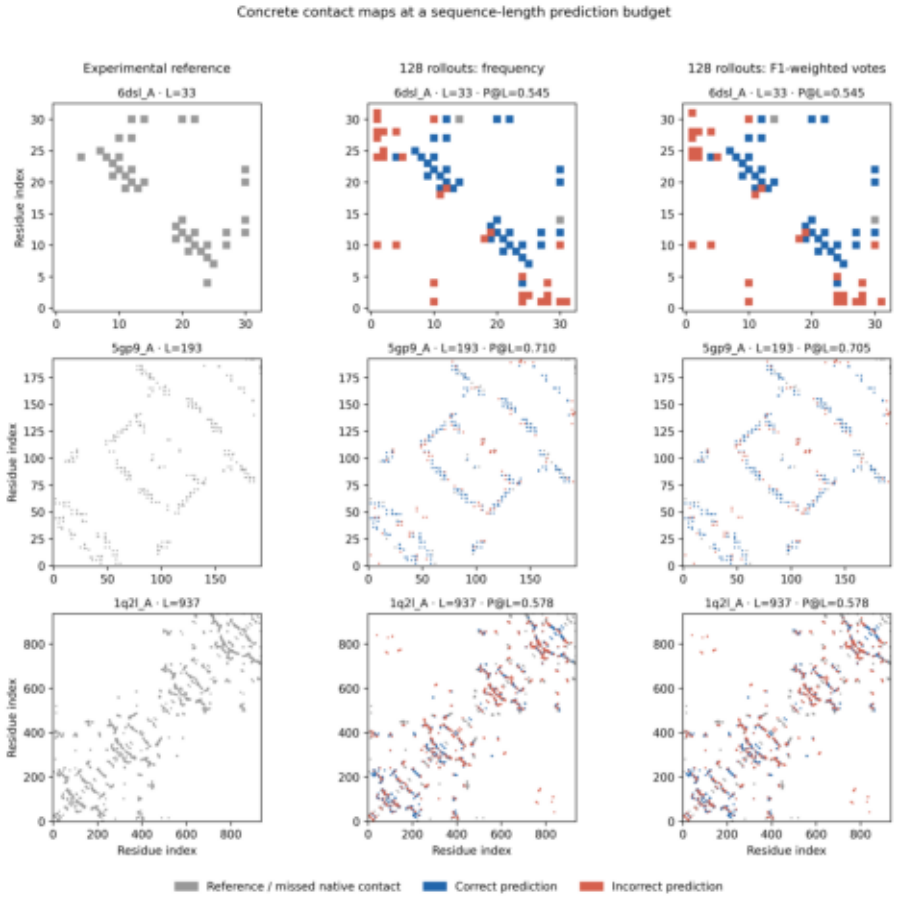
Limits and remaining confirmation

The original guided cohort must finish before final selection. The reserved comparison will freeze exactly two configurations and require all 1,473 structures to finish before metric analysis. The reserved set was previously used for the proofreader release; it is not an untouched external benchmark. Generator pretraining decontamination is not established.

Adaptive additions and unadjusted validation intervals are identified explicitly. Weighted votes are ranking scores, not calibrated pair probabilities. A positive local proposal-ranking effect need not improve a complete map, and good calibration on ordinary rollouts need not survive guided distribution shift. The remaining experiment will determine whether the original search policies improve the measured quality/cost frontier.

contact_examples.png

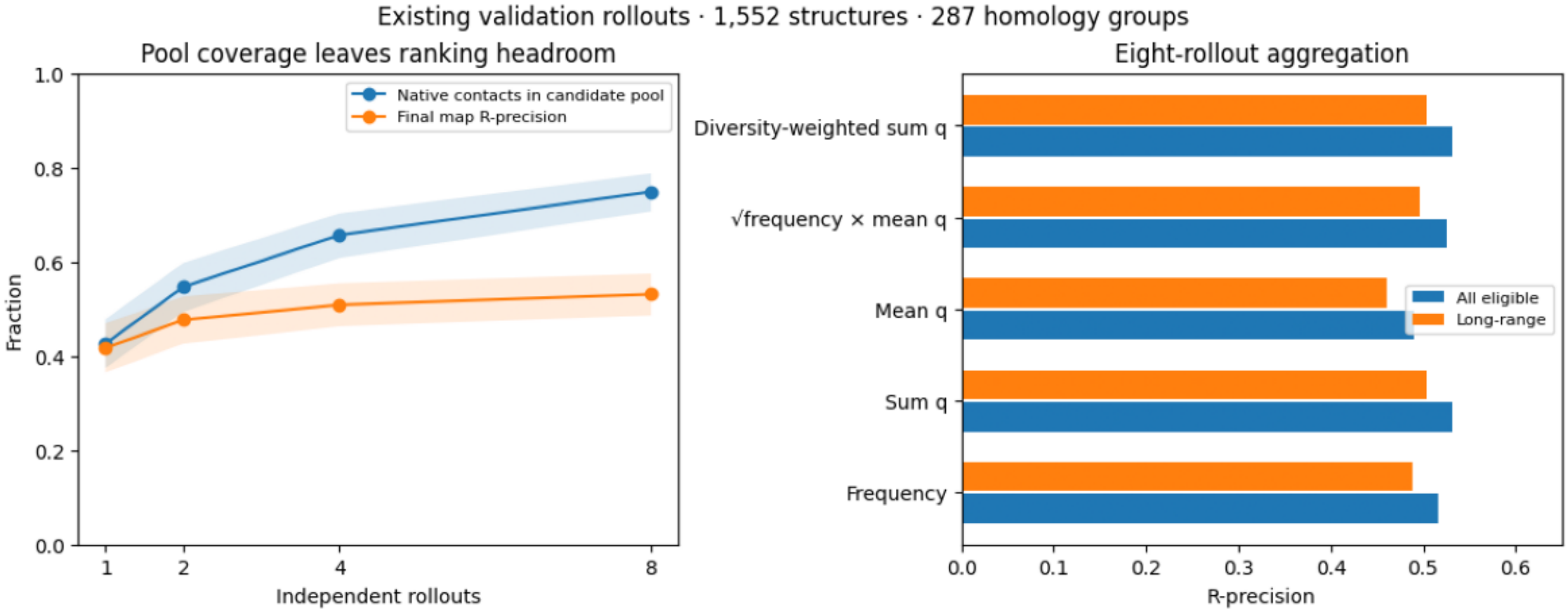
Three validation proteins chosen as the first, middle and last length ranks for earlier numerical checks, without accuracy selection. Both methods rank the same 128 H100 rollouts and predict L unique pairs. Matrices are mirrored only for visualization; each pair counts once. Gray native contacts remain visible where not selected. Point markers are enlarged for legibility on long sequences. These examples are descriptive and do not estimate an average effect. References use the inherited exp222 native-only pyconfind definition, with sequence separation at least six. The published coordinate CSV reproduces this figure.




```
$ plot_contact_examples.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/contact_examples.csv.gz --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/contact_examples.png
```

existing_validation.png

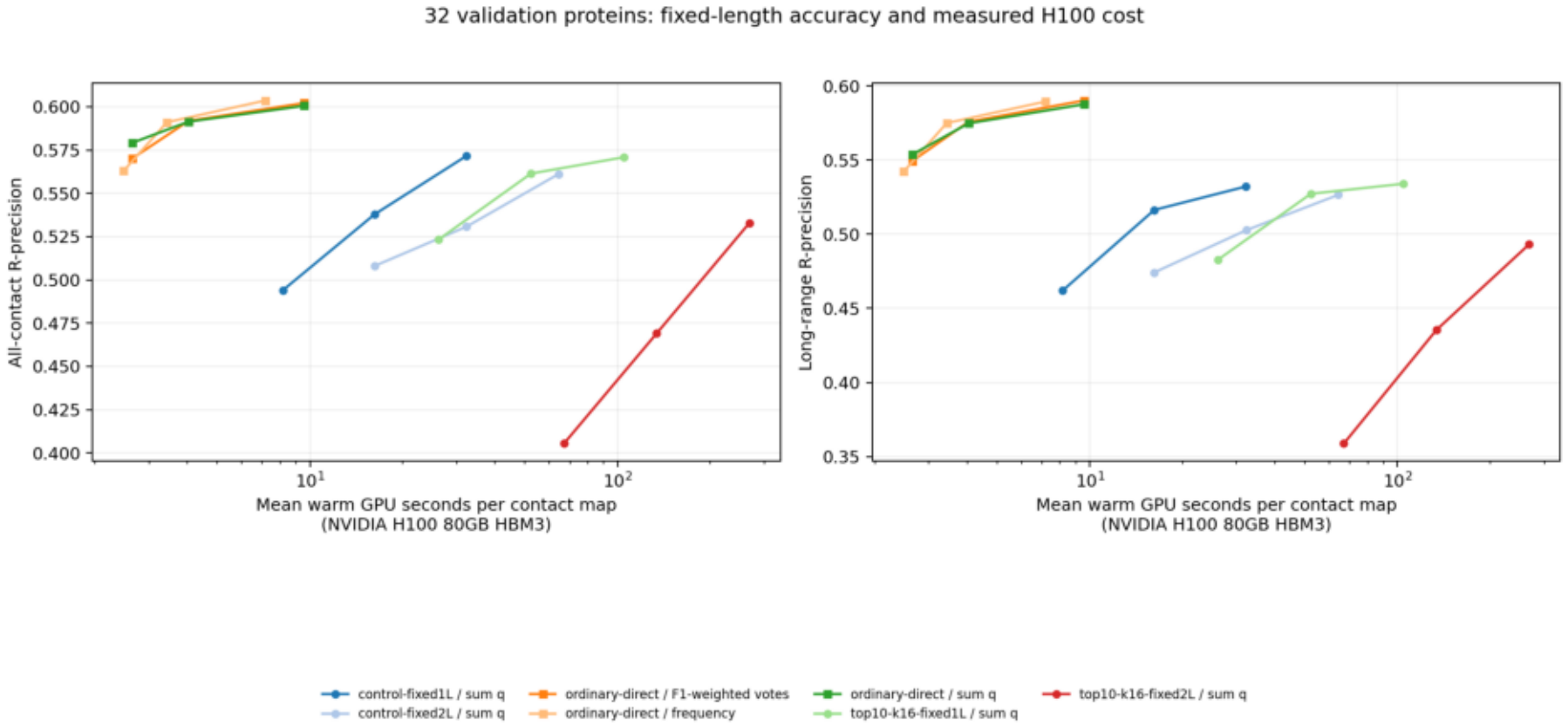
Validation-only reanalysis. Bands resample homology groups (95% intervals). Oracle coverage uses labels only to bound achievable recall. The small number of structures with fewer than eight usable rollouts retains all available samples. Long-range means separation ≥24.



$ plot_existing.py

length_probe_frontier.png

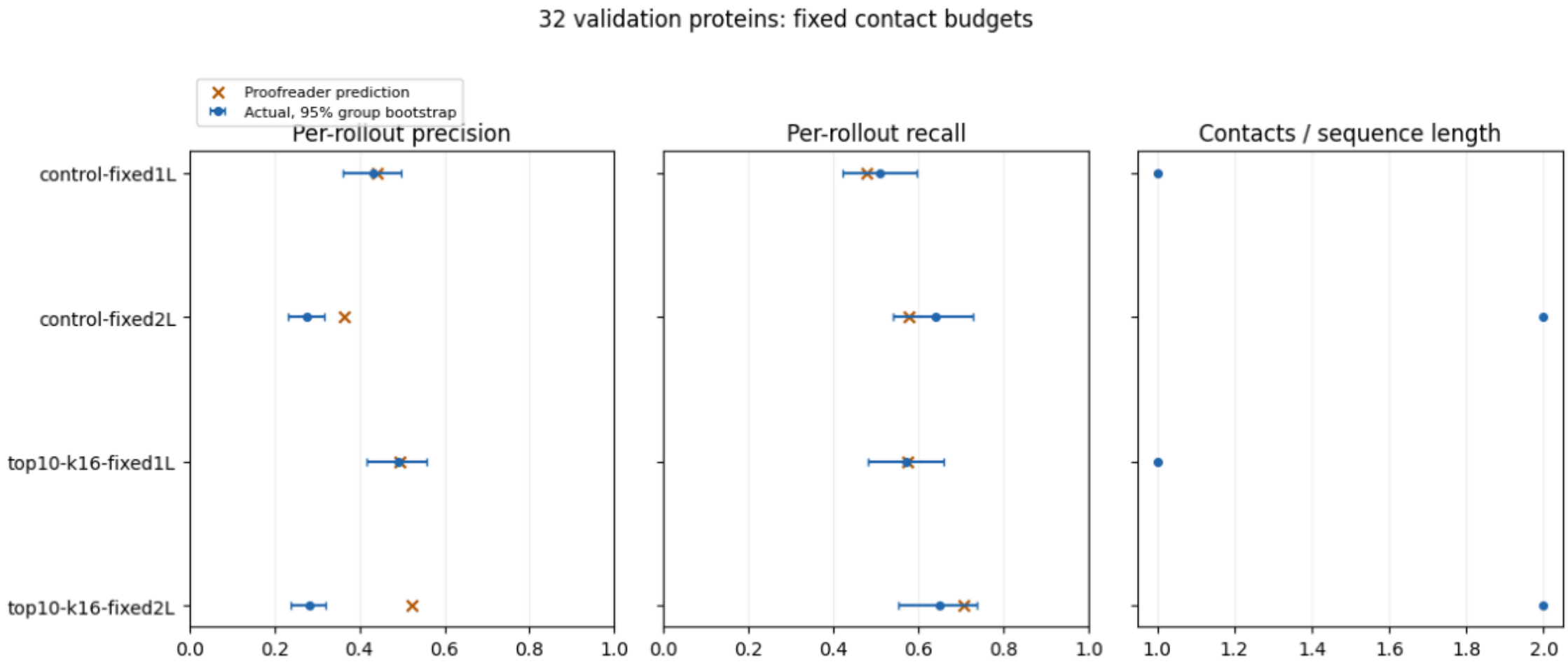
GPU model: NVIDIA H100 80GB HBM3. Each point averages the same planned structures. Guided curves use sum-q aggregation over increasing independent trajectories; ordinary curves show frequency, sum q and predicted-rollout-F1-weighted votes. The F1 rule is an exploratory addition after the ordinary-pool pilot. Search cost includes discarded proposals, lookahead, population branches, repair and final scoring. Ordinary frequency omits proofreader cost. GPU resource seconds sum the work for all trajectories; they are not parallel serving latency. Model staging and loading are excluded. These are measured compute envelopes, not exactly cost-matched experiments.




```
$ plot_frontier.py --input experiments/exp368_evals_proofreader_guided_contact_maps/data/length_probe/per_protein.csv.gz \
  experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_direct/per_protein.csv.gz --plan experiments/exp368_evals_proofreader_guided_contact_maps/length_probe.json \
  --out experiments/exp368_evals_proofreader_guided_contact_maps/plots/length_probe_frontier.png --title '32 validation proteins: fixed-length accuracy and measured H100 cost'
```

length_probe_quality.png

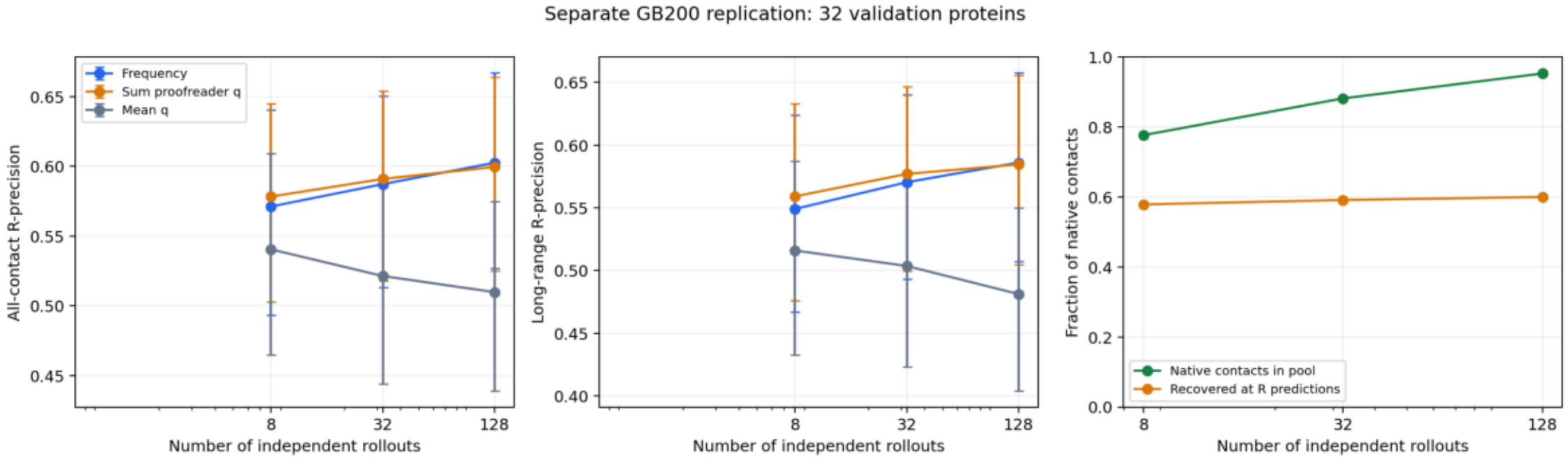
32 proteins, equally weighted after averaging trajectories within each protein. Blue intervals resample homology groups. Orange crosses show mean critic predictions. Fixed-length L and 2L maps are nested prefixes of the same continuations.




```
$ plot_trajectory_quality.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/length_probe --plan \
  experiments/exp368_evals_proofreader_guided_contact_maps/length_probe.json --out experiments/exp368_evals_proofreader_guided_contact_maps/plots/length_probe_quality.png --table \
  experiments/exp368_evals_proofreader_guided_contact_maps/data/length_probe/trajectory_summary.csv --title '32 validation proteins: fixed contact budgets'
```

pilot_direct_gb200.png

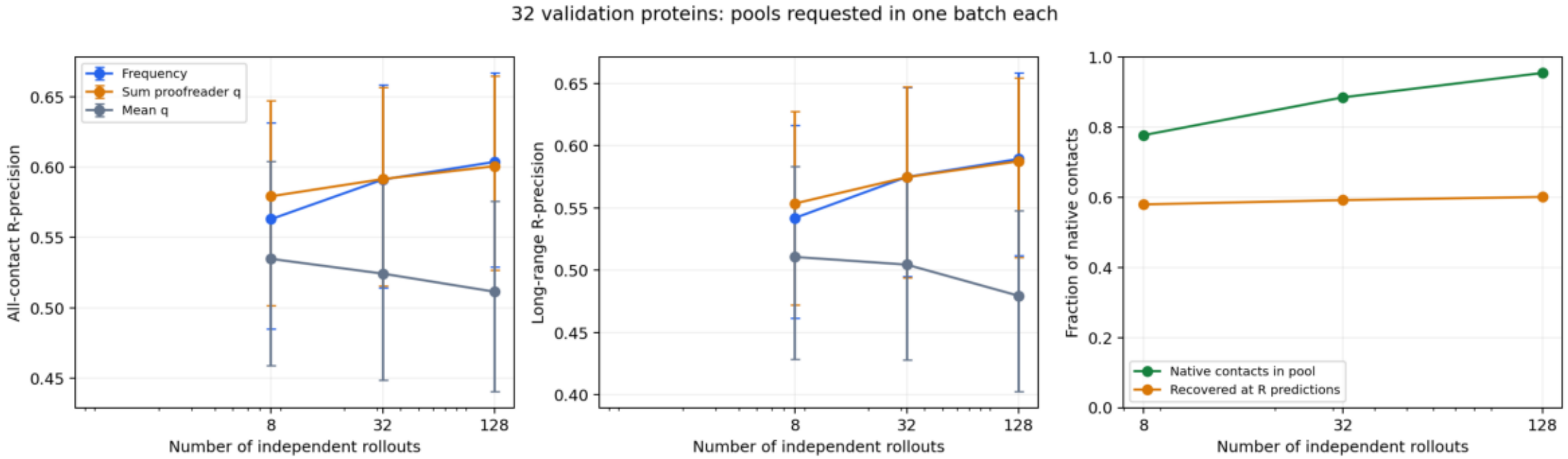
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_direct_gb200 --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_direct_gb200.png --title 'Separate GB200 replication: 32 validation proteins'
```

pilot_direct_pool.png

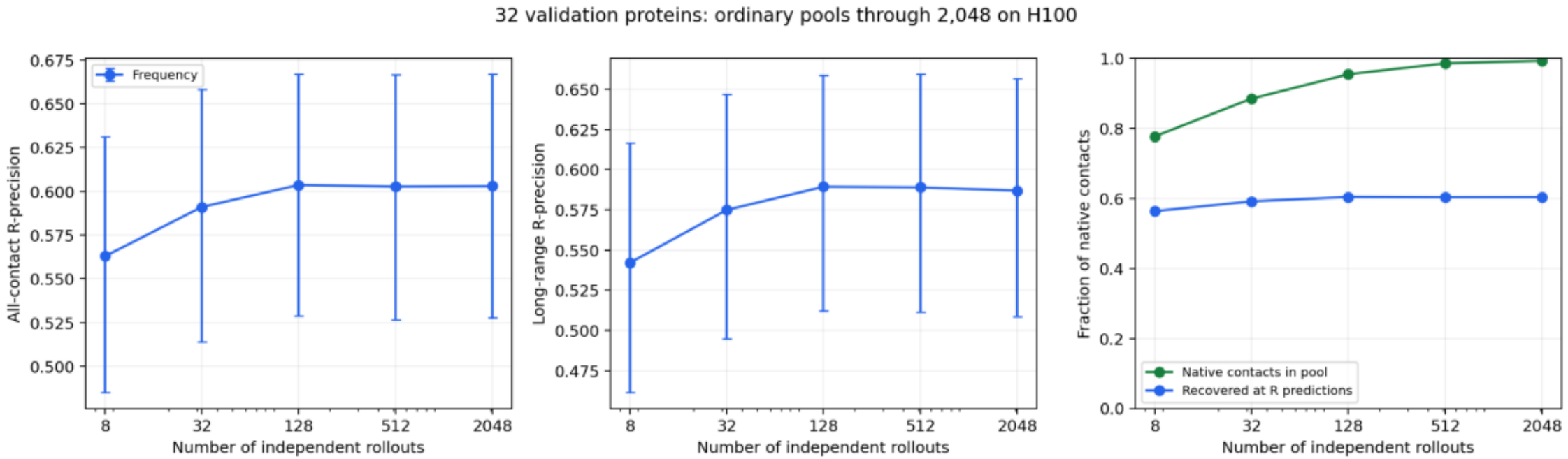
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_direct --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_direct_pool.png --title '32 validation proteins: pools requested in one batch each'
```

pilot_large.png

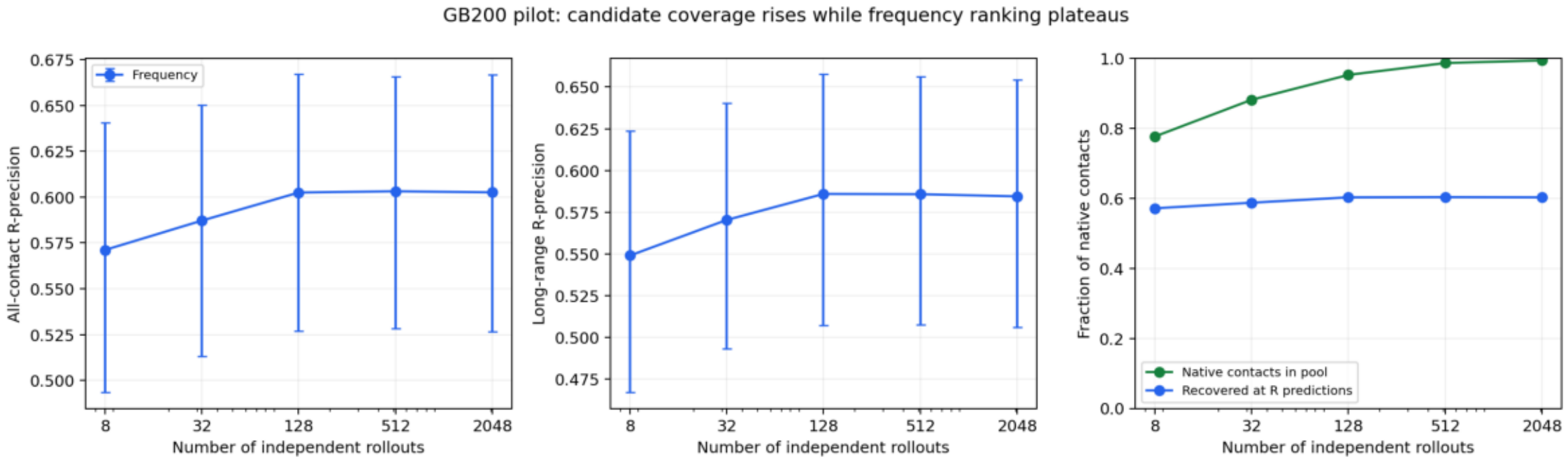
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_pool_scaling --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_large.png --title '32 validation proteins: ordinary pools through 2,048 on H100' --methods frequency \
  --coverage-method frequency
```

pilot_large_gb200.png

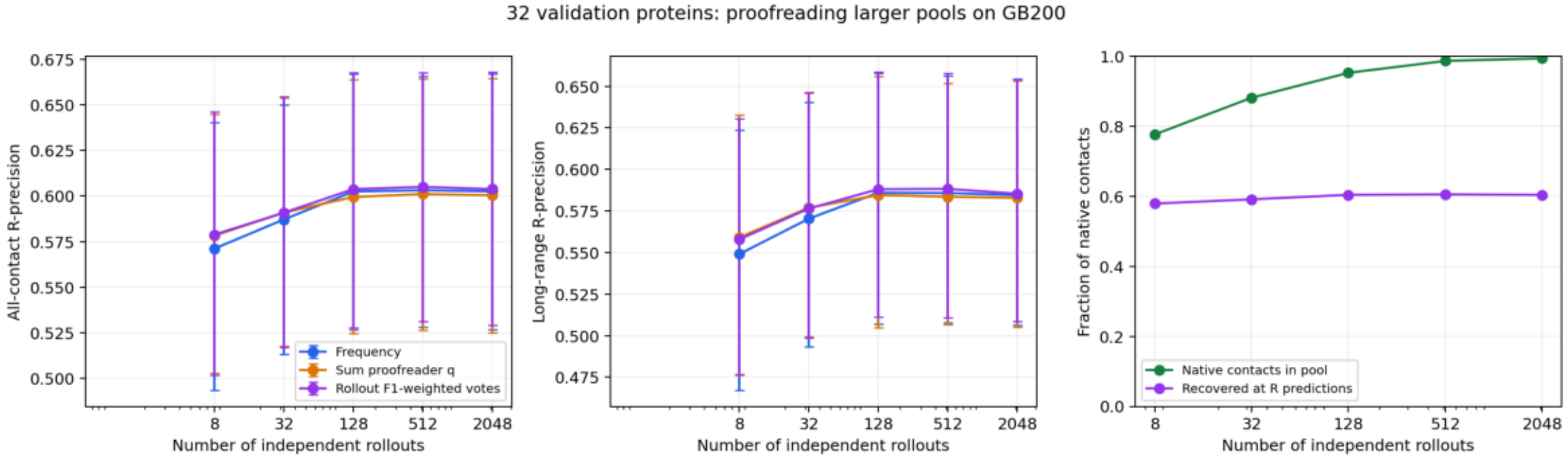
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_pool_scaling_gb200 --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_large_gb200.png --methods frequency --coverage-method frequency --title \
  'GB200 pilot: candidate coverage rises while frequency ranking plateaus'
```

pilot_large_scored_gb200.png

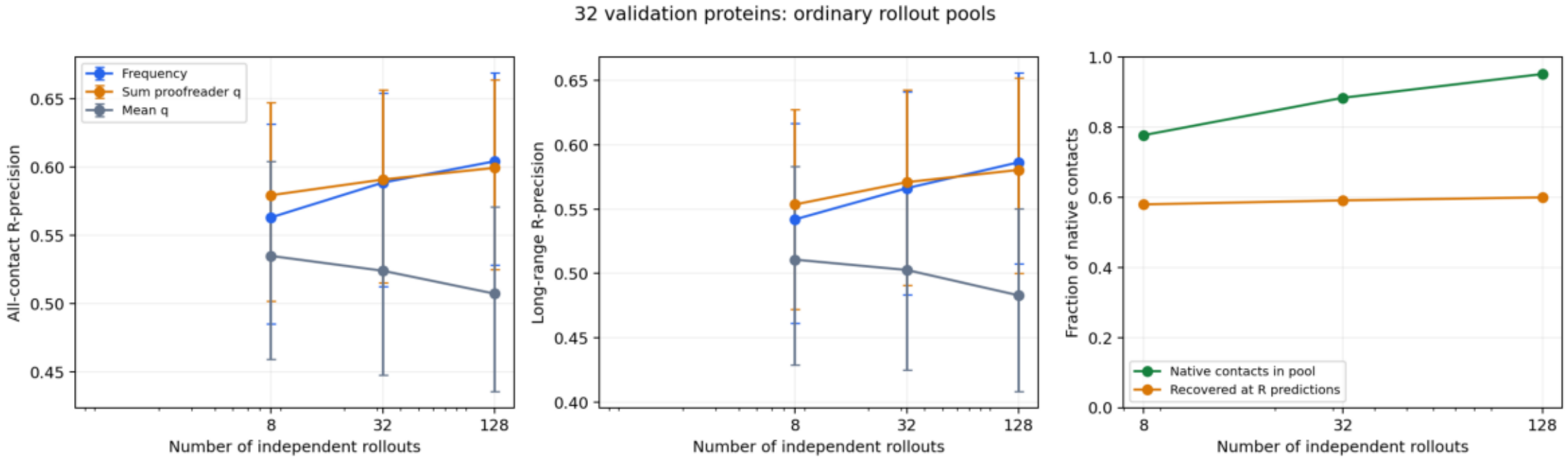
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_pool_scaling_scored_gb200 --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_large_scored_gb200.png --title '32 validation proteins: proofreading larger pools on GB200' --methods \
  frequency sumq f1_weighted_frequency --coverage-method f1_weighted_frequency
```

pilot_pool.png

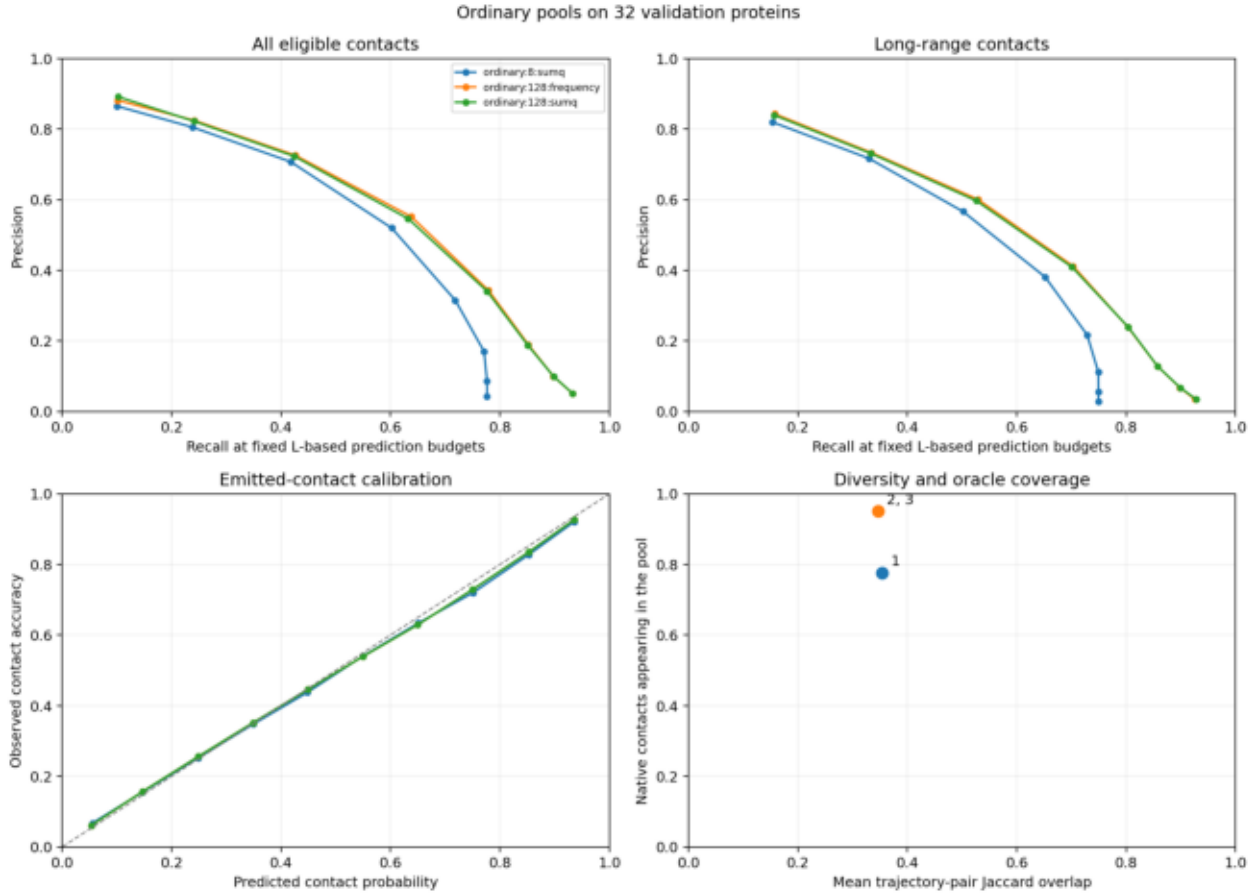
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_pool --out experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_pool.png
```

pilot_pool_diagnostics.png

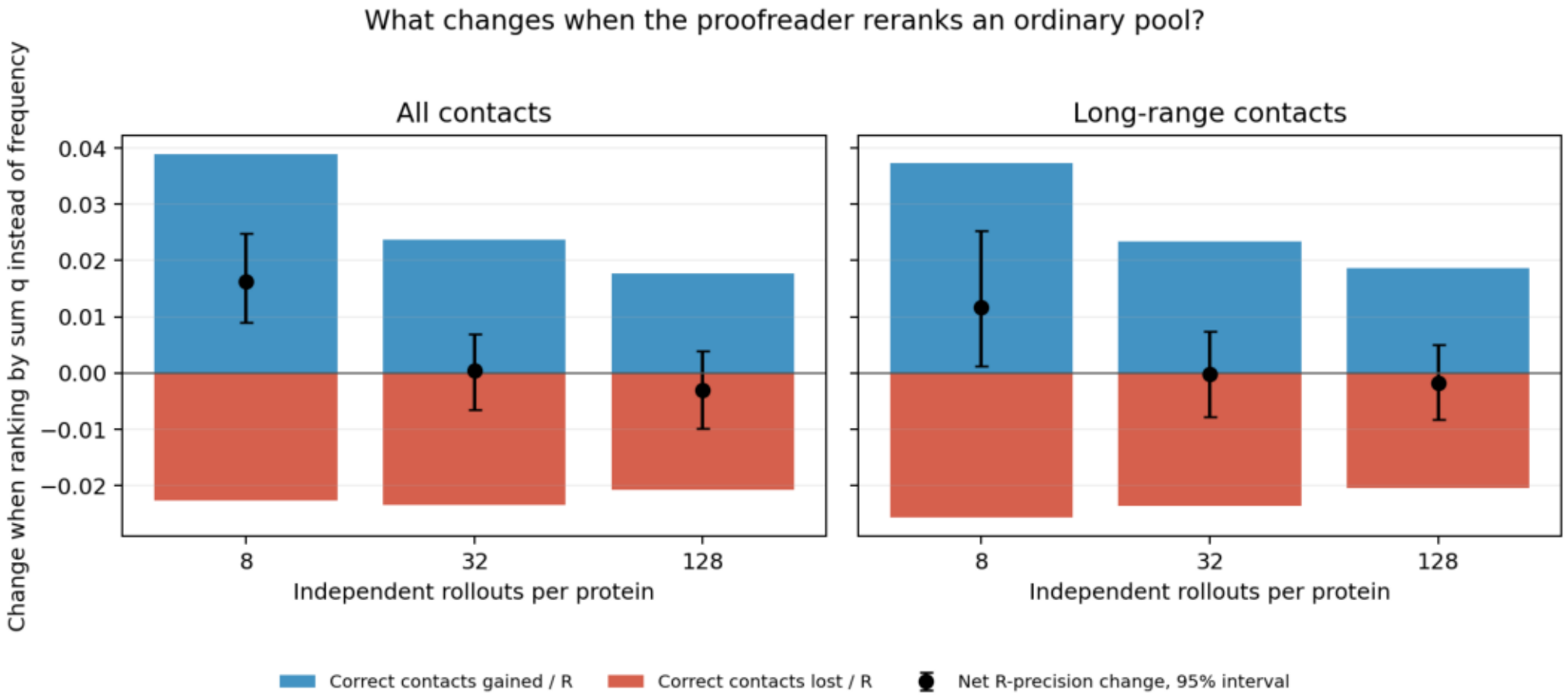
Curves average over the same structures at budgets 0.1, 0.25, 0.5, 1, 2, 4, 8 and 16 times sequence length. Unfilled slots count as misses; curves do not use native contact counts to choose the prediction budget. Calibration pools first occurrences within each rollout and is contact-weighted. Diversity and coverage are structure means; at most 128 deterministically sampled trajectory pairs measure overlap. Numbers in the coverage panel follow legend order; multiple aggregators of the same pool share calibration and diversity.




```
$ plot_diagnostics.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_pool/diagnostics --spec ordinary:8:sumq ordinary:128:frequency ordinary:128:sumq \
  --title 'Ordinary pools on 32 validation proteins' --out experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_pool_diagnostics.png
```

pilot_reranking.png

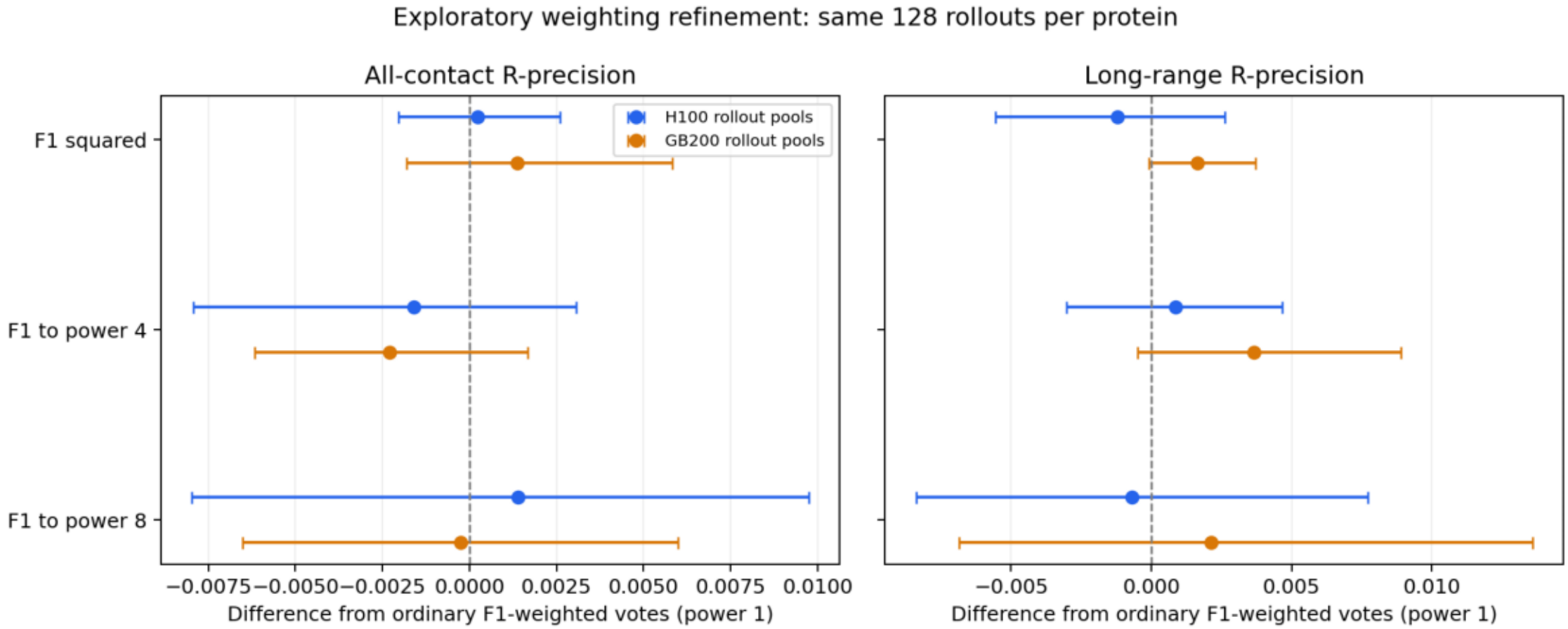
Both rankings use exactly the same sampled candidate union and deterministic label-independent tie breaks. Bars count true contacts entering and leaving the top R; their sum equals the observed R-precision change, verified for every input. Proteins are weighted equally; intervals resample homology groups. R uses native contact count only for retrospective evaluation. This is an adaptive diagnostic, not a new policy.




```
$ analyze_reranking.py --root experiments/exp368_evals_proofreader_guided_contact_maps/_cache/pilot-direct-scored --plan \
  experiments/exp368_evals_proofreader_guided_contact_maps/pilot_direct.json --targets experiments/exp356_models_pdb_rollout_proofreader/_cache/prepared-v2/targets.parquet --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_direct --plot experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_reranking.png
```

pilot_weight_power.png

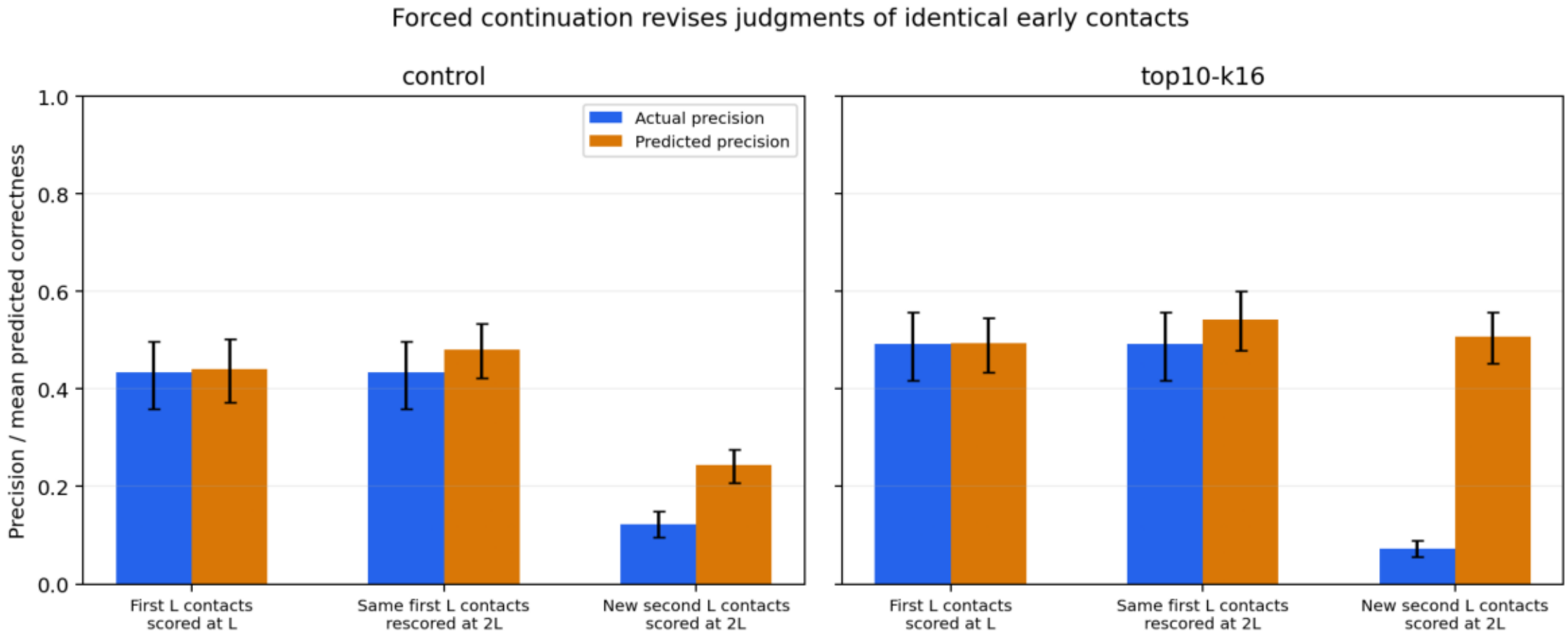
Paired 95% homology-group bootstrap intervals on the complete 32-protein pilot. The same proteins are used on both hardware/runtime phases; they are not independent biological cohorts. Powers 2, 4, 8 were fixed after observing the power-1 validation result, before computing these comparisons. All variants reuse identical generated pools and reader outputs within each runtime. Intervals are exploratory and unadjusted for multiple comparisons; no test metrics are used.




```
$ plot_weight_power.py --h100 experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_weight_power --gb200 \
  experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_weight_power_gb200 --out experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_weight_power.png
```

prefix_revision.png

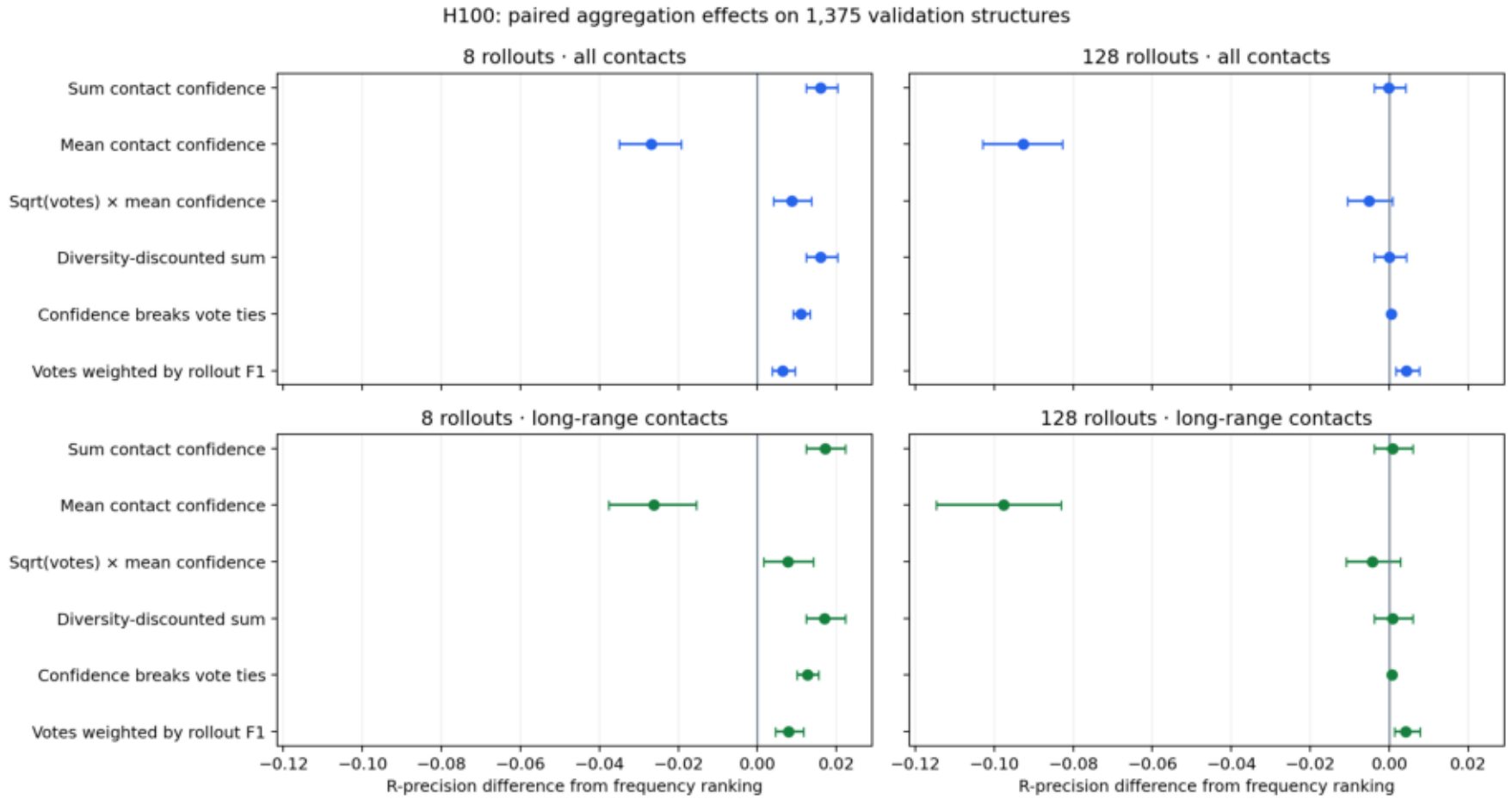
Retrospective analysis of all 32 validation proteins and four trajectories per sampler. First-L pairs and labels are verified identical before and after continuation. Means average trajectories within each protein, then proteins; 95% intervals resample homology groups. The second half is generated with EOS disabled. No test-set data or new model inference is used.




```
$ analyze_prefix_revision.py --plan experiments/exp368_evals_proofreader_guided_contact_maps/length_probe.json --root \
  experiments/exp368_evals_proofreader_guided_contact_maps/_cache/length-probe-native-v1 --out experiments/exp368_evals_proofreader_guided_contact_maps/data/prefix_revision --plot \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/prefix_revision.png
```

validation_aggregation_effects.png

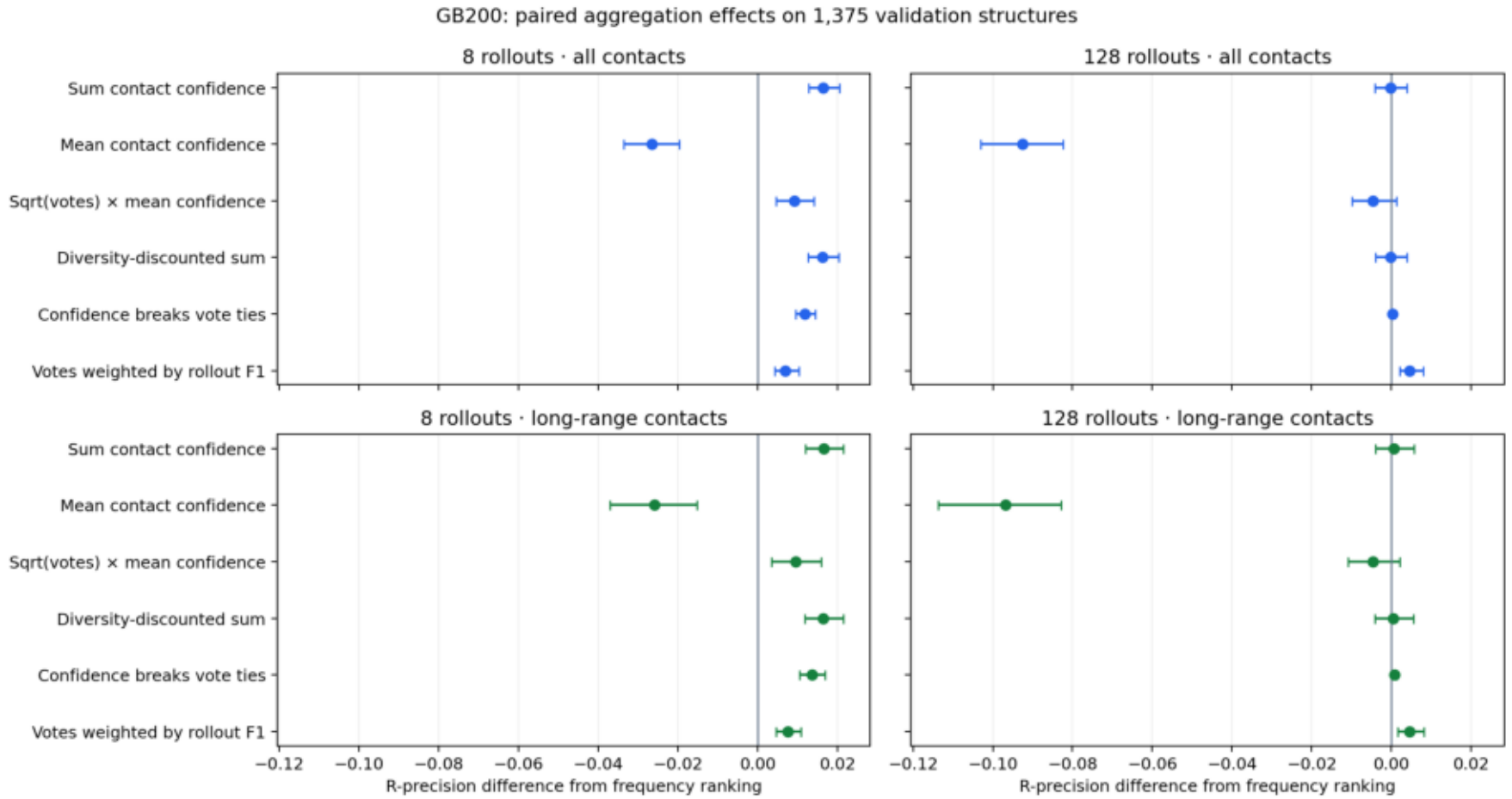
Paired differences on identical generated pools. Error bars are 95% homology-group bootstrap intervals, without multiplicity adjustment. All planned structures are included; long-range metrics exclude structures with no native contact at sequence separation ≥24. F1 weighting and confidence tie-breaking were added after the ordinary-pool pilot and remain exploratory. No new reserved-test scores are shown.




```
$ plot_aggregation_effects.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_pool --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_aggregation_effects.png --title 'H100: paired aggregation effects on 1,375 validation structures'
```

validation_aggregation_effects_gb200.png

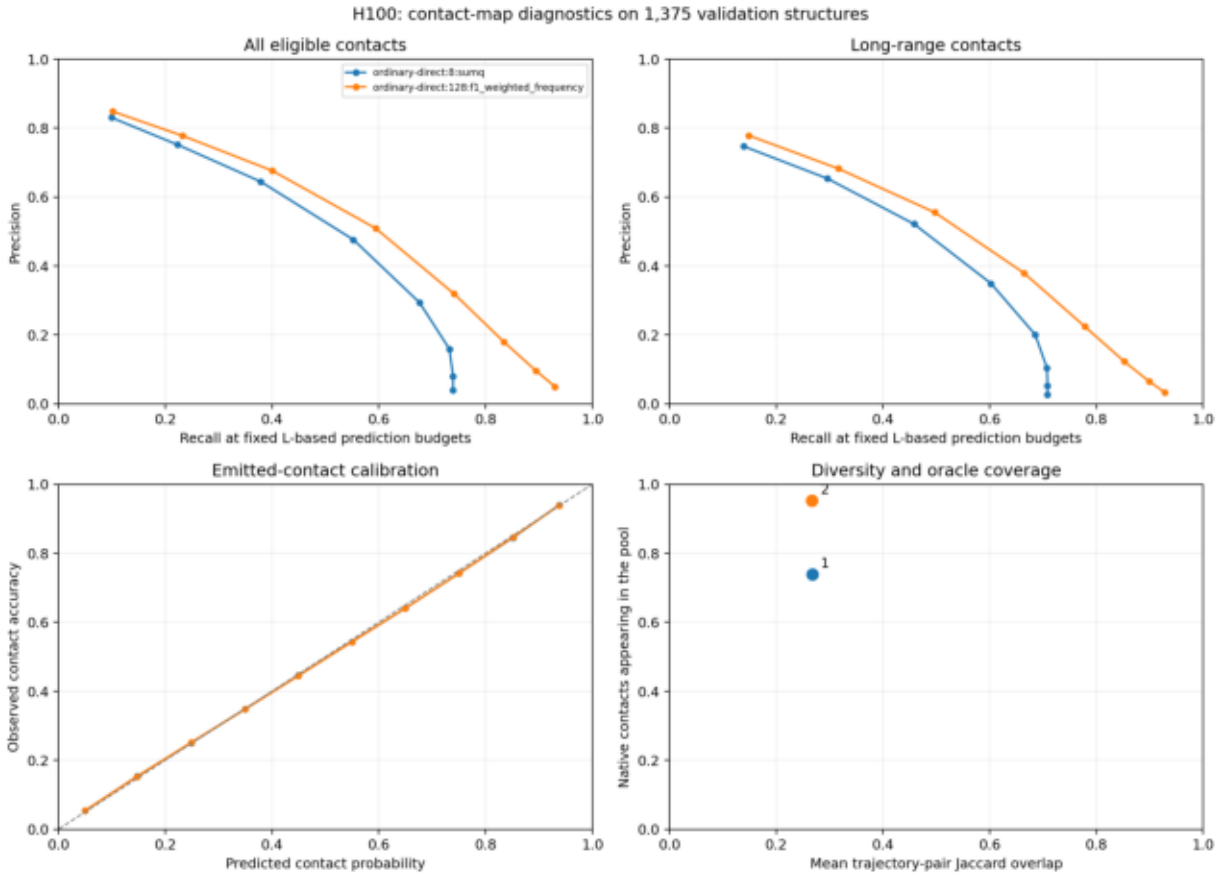
Paired differences on identical generated pools. Error bars are 95% homology-group bootstrap intervals, without multiplicity adjustment. All planned structures are included; long-range metrics exclude structures with no native contact at sequence separation ≥24. F1 weighting and confidence tie-breaking were added after the ordinary-pool pilot and remain exploratory. No new reserved-test scores are shown.




```
$ plot_aggregation_effects.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_pool_gb200 --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_aggregation_effects_gb200.png --title 'GB200: paired aggregation effects on 1,375 validation structures'
```

validation_diagnostics.png

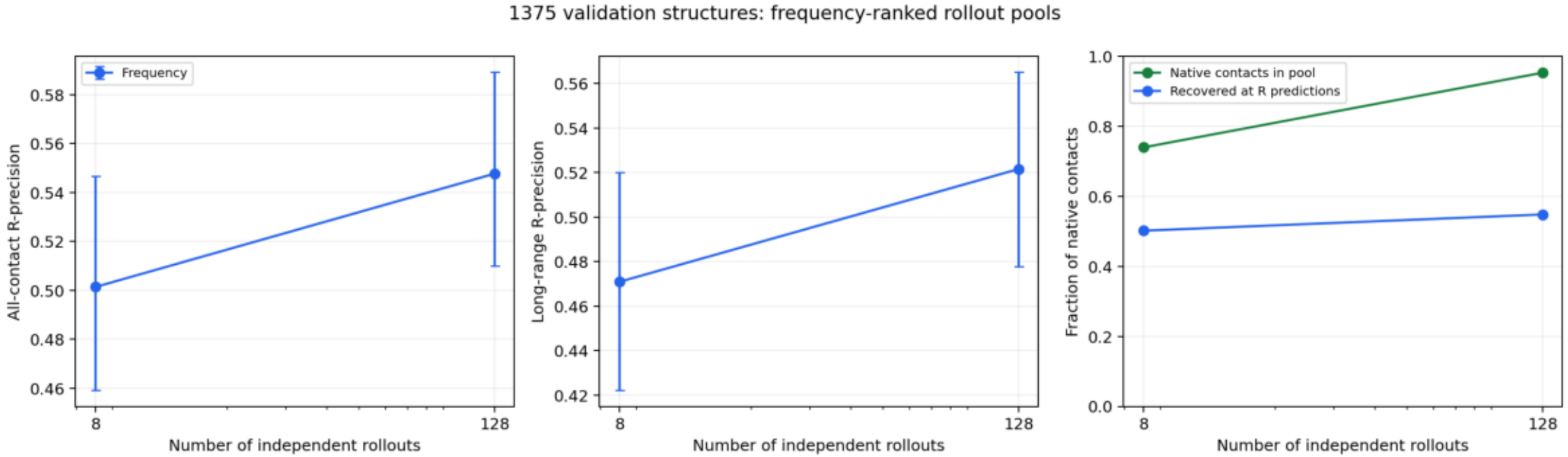
Curves average over the same structures at budgets 0.1, 0.25, 0.5, 1, 2, 4, 8 and 16 times sequence length. Unfilled slots count as misses; curves do not use native contact counts to choose the prediction budget. Calibration pools first occurrences within each rollout and is contact-weighted. Diversity and coverage are structure means; at most 128 deterministically sampled trajectory pairs measure overlap. Numbers in the coverage panel follow legend order; multiple aggregators of the same pool share calibration and diversity.




```
$ plot_diagnostics.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_diagnostics --spec ordinary-direct:8:sumq \
  ordinary-direct:128:f1_weighted_frequency --out experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_diagnostics.png --title \
  'H100: contact-map diagnostics on 1,375 validation structures'
```

validation_frequency.png

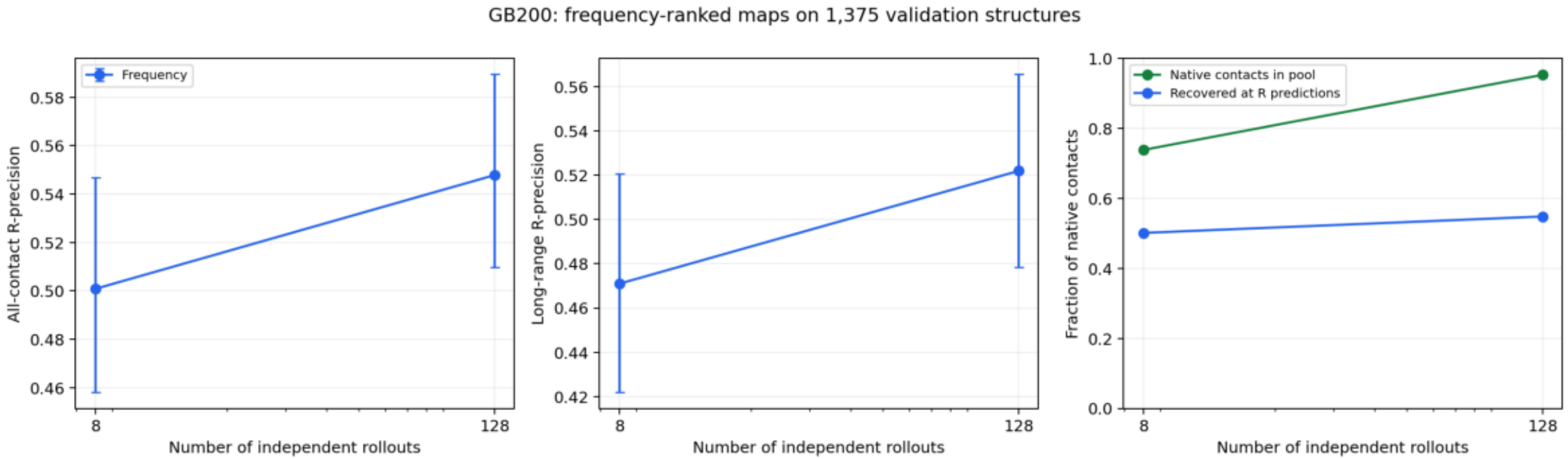
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_frequency --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_frequency.png --title '1375 validation structures: frequency-ranked rollout pools' --methods frequency \
  --coverage-method frequency
```

validation_frequency_gb200.png

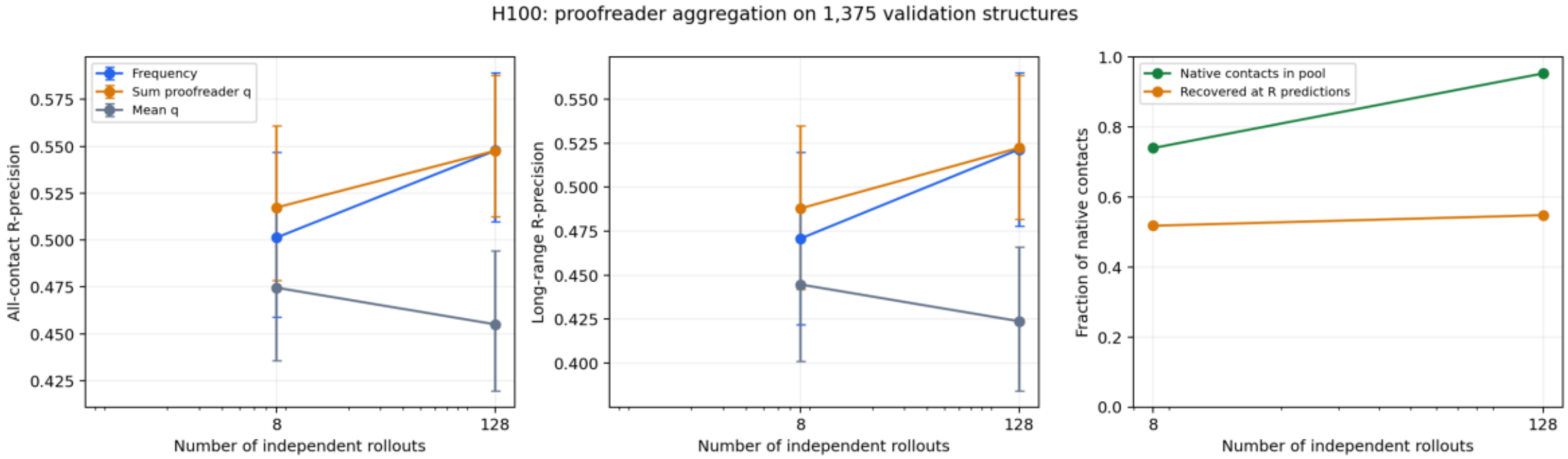
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_frequency_gb200 --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_frequency_gb200.png --title 'GB200: frequency-ranked maps on 1,375 validation structures' --methods \
  frequency --coverage-method frequency
```

validation_pool.png

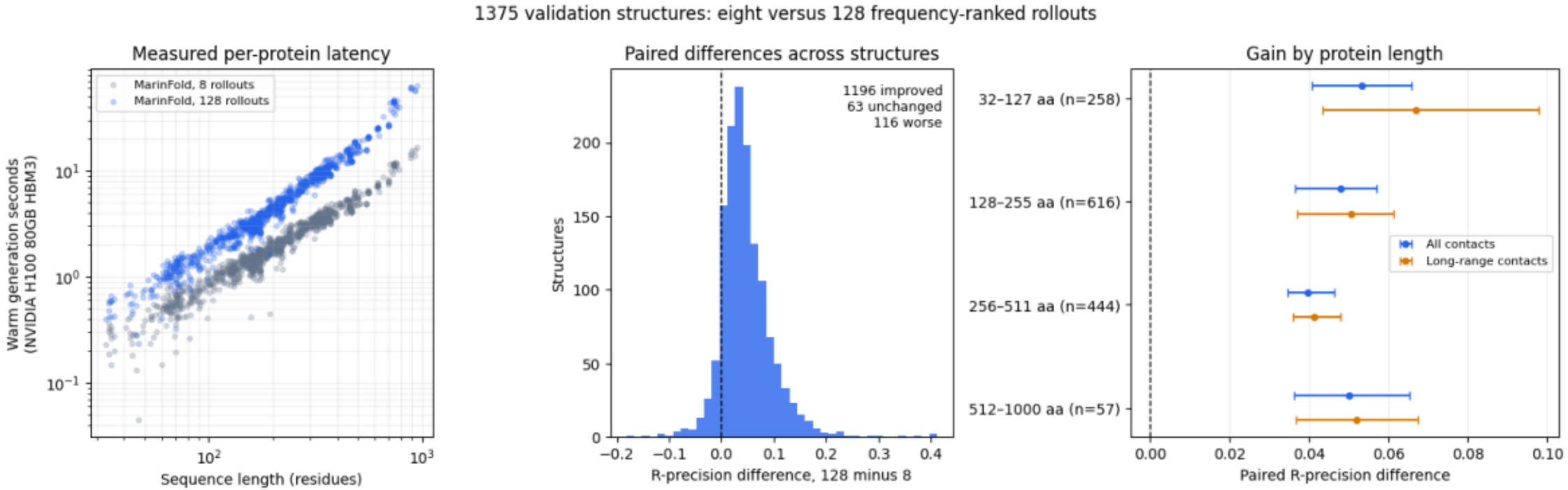
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_pool --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_pool.png --title 'H100: proofreader aggregation on 1,375 validation structures'
```

validation_pool_scaling.png

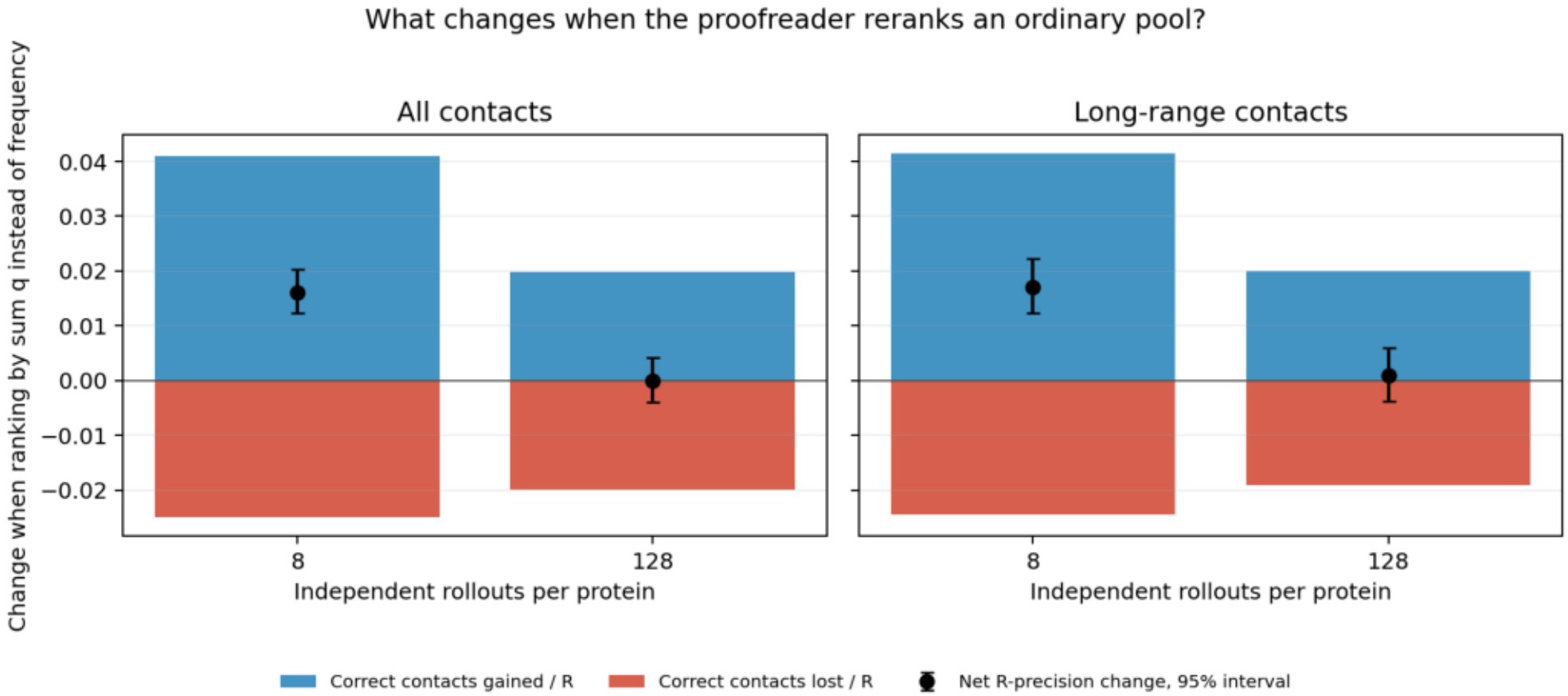
Each latency point is one (protein, pool-size) inference, excluding model load and artifact writes. Length strata use fixed boundaries at 32, 128, 256, 512 and 1001 residues. Intervals resample related groups within each stratum and describe paired differences. Strata are descriptive; they do not select proteins or tune a test policy.




```
$ plot_pool_scaling.py --input experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_frequency/per_protein.csv.gz --plan \
  experiments/exp368_evals_proofreader_guided_contact_maps/validation_pool.json --data-out experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_frequency --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_pool_scaling.png
```

validation_reranking.png

Both rankings use exactly the same sampled candidate union and deterministic label-independent tie breaks. Bars count true contacts entering and leaving the top R; their sum equals the observed R-precision change, verified for every input. Proteins are weighted equally; intervals resample homology groups. R uses native contact count only for retrospective evaluation. This is an adaptive diagnostic, not a new policy.




```
$ analyze_reranking.py --plan experiments/exp368_evals_proofreader_guided_contact_maps/validation_pool.json --targets \
  experiments/exp356_models_pdb_rollout_proofreader/_cache/prepared-v2/targets.parquet --root \
  experiments/exp368_evals_proofreader_guided_contact_maps/_cache/validation-pool-scored-native-v1 --out \
  experiments/exp368_evals_proofreader_guided_contact_maps/data/validation_pool --plot experiments/exp368_evals_proofreader_guided_contact_maps/plots/validation_reranking.png
```